A_GMX on crowded CCDs

background override, serial-register streaks, muon line components, topology split, crossing split and track linking

comm_scripts clustering study

code of 29 October 2026 (`clustering_tests/crowded.py`)

Abstract

On CCDs where tracks cover a large part of the image (surface runs such as L2 ACM101, the Am-241 test CCD), A_GMX had three failures: its row-by-row background fit picked the wrong whole-electron pedestal in rows with $\sim 10\,e^-$/pix of dark charge; serial-register streaks were merged into tracks; and long straight tracks (muons) were cut into several clusters wherever they crossed other tracks, while touching tracks were merged into star-shaped clusters of up to 10^4 px. *Crowded mode* fixes these and is switched on per image when the image is crowded. The straight tracks are found by a separate finder, but only their *axis* is passed to A_GMX as a hint: the cluster's width, footprint and the charge shared with crossing tracks remain A_GMX's own. On L2, injected muons go from 4.1 fragments and 45 % of their charge in one cluster to 1.9 fragments and 93 %; Fe-55 Kα narrows from 220 to 85 eV on the worst CCD and gains 14–65 % events. On the Am-241 crowded CCD (EXT4) the energy scale of the constrained 10–30 keV fit is unchanged ($k = 1.0004$ vs 1.0009 on the same images), and on EXT3 too ($k = 0.9952 \pm 0.0008$ vs 0.9964 ± 0.0010, 29 images) once the fit is started from narrow line widths (section 3.5). Crowded mode is the default (`crowded='auto'`): it runs only on images that pass the trigger, and other images are clustered as before.

Since 29 October crowded mode goes one step further by default (`crowded_shapes='on'`): each straight track is a *line component* of the mixture (charge per length and width fitted, section 2.4) instead of a group of clouds, and every cluster gets a *topology split* (one event has a simple skeleton, section 2.5). The charge of touching objects in a muon's cluster drops from 28 % to 8.5 % (median) and the straight tracks' energy loss now peaks at 0.27 keV/µm, the most probable value for a minimum-ionising particle (figure 9).

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1 When is an image crowded?

The trigger is the *core fill*: the fraction of active pixels whose charge, smoothed with a Gaussian of 1 px, exceeds $\max(60\,e^-,\, 5\,\sigma_\lambda)$, where $\sigma_\lambda = \sqrt{\lambda}/(2\sqrt{\pi})$ is the noise of the smoothed dark charge λ. It measures how much of the CCD is covered by deposits, independent of the background level.

image type	core fill	crowded mode with `auto`
L2 ACM101 surface run (EXT1/2/4)	0.27–0.31	on
Am-241 AMTest EXT3 / EXT4	0.27 / 0.43–0.67	on
Fe-55 production CCDs (gn_lowcharge_study)	0.015–0.02	off

`crowded='auto'` switches crowded mode on for core fill ≥ 0.10; `'on'` and `'off'` force it. Below the threshold nothing in this note runs, and A_GMX output is identical to the code without crowded mode (unit test).

2 The five pieces

2.1 Background override (only at high dark charge)

A_GMX's background model fits each row with a Poisson⊗Gaussian (offset o, dark charge λ, noise s). With $\lambda \gtrsim 5\,e^-$/pix the zero-electron peak is gone and o is only known modulo whole electrons: on L2 EXT1, 395 of 500 rows took a pedestal 4–85 e^- too high, which made energies wrong by up to $-6\,e^-$/pix × the cluster area and produced a 10–160k px "blob". The override:

1. is applied only for electron-counting images (median $s < 0.45\,e^-$) with median $\lambda \geq 3\,e^-$/pix;
2. flags a row as bad when more than 0.5 % of its pixels lie below $o - 0.5 - 3s$, and gives it the whole-electron pedestal of the good rows (keeping its fractional part);
3. re-estimates λ in cells of 10×64 px by a truncated-Poisson fit below the 90 % quantile (so that the halos of tracks do not raise it), and overrides the row model where the two differ by more than $\max(0.3,\, 5\sigma)\,e^-$/pix; bad rows take the cell value.

The gate on λ matters: on Am-241 (λ = 0.2–2.4 e^-/pix) the dense X-ray field fools both tests (EXT3: 40–70 % of the cells changed, energies −0.2 %, 59.5-keV events lost; EXT4: 429 "bad" rows and backgrounds of $10^4\,e^-$/pix). A unit test checks that the override never moves a row's pedestal away from the truth.

Hot-column bands. On L2 9 EXT1 columns 893–925 carry $+8\,e^-$/pix of extra dark charge with sharp edges; the 64-column cells cannot follow them and A_GMX turned the band into 700–3000 px faint "blobs". For every column and block of 32 rows the mean of the free pixels (deposits above 8σ masked) minus the model is measured, smoothed along the column only, and added to λ where it is positive, $\geq \max(2\,e^-, 4\sigma)$ and part of a run of ≥ 4 adjacent columns. Coherent *deficits* are not applied: they sit in rows with undershoot after bright tracks, where the low-count statistics still fit the row model. Fe-55 on L2 is unchanged (EXT1 gains 3 % events).

Glow flag. Clusters of ≥ 100 px whose mean charge per pixel is below $3\sqrt{\lambda}$ are background structure the model does not follow: class `glow` (`agmx_cls` = 11), 5–12 per L2 CCD.

2.2 Serial-register streaks

Charge clocked into the serial register lands in one row with no vertical spread. A significant pixel is *single-row* when the pixels above and below hold < 25 % of its charge; horizontal runs of ≥ 8 such pixels (1-px gaps bridged; pieces ≤ 10 px apart in a row joined) are removed from the image before clustering and reported as clusters with class `artifact-row` and regime `serial` (`agmx_regime` = 3).

2.3 Straight-track hints

Finding. The *core* (as in section 1) is reduced to its 1-px skeleton; a Hough transform in tiles of 160×160 px (half overlapping; votes summed over 3 neighbouring ρ bins so that a wobbling skeleton still counts) gives candidate lines with ≥ 40 votes. Lines within 10° of a row are left to the streak finder. Each candidate is refined (charge-weighted PCA of the core pixels within 2 px) and walked along the image in both directions. A step is occupied when a core pixel within ± 1 px of the axis is a transverse ridge (its smoothed charge at least twice the smaller of the two values at ± 3.5 px) or lies on a track already accepted (so crossings do not stop the walk); the walk ends after 6 empty steps.

Accepting. A line is a straight track if it is ≥ 40 px long, ≥ 85 % occupied, its median transverse rms is ≤ 2.2 px, it is not a duplicate, and its charge is *continuous*: of the steps along the axis, at most 15 % may lie below half and at most 8 % below 0.3 of the median axis charge. The continuity test separates muons (injected: 90th percentile 0.09) from chains of X-ray blobs in a dense field (Am-241: 10th percentile 0.17); without it the finder took 166 "tracks" on an Am-241 image, with it 13 (figure 7).

Using the hints. Nothing is removed from the image. A_GMX runs as always (seeds, EM of the Gaussian clouds, merge graph, island arbitration). Then, for each hint (axis centre $\mathbf{c}$, direction $\mathbf{u}$, extent $t_0..t_1$, rms width w), every cloud k with

$$|(\boldsymbol{\mu}_k - \mathbf{c}) \cdot \mathbf{n}| \leq \max(1.5,\, w), \qquad t_0 - 3 \leq (\boldsymbol{\mu}_k - \mathbf{c}) \cdot \mathbf{u} \leq t_1 + 3$$

($\mathbf{n} \perp \mathbf{u}$) goes into one cluster, unless it is elongated (axis ratio ≥ 1.6) at more than 30° to the hint, which makes it part of a crossing track. The cluster gets class `muon` and `straight_track = True`. Because the pixels' charge is still split by the mixture's responsibilities, a crossing track keeps its charge and stays one cluster, and the muon's width is the width A_GMX fits (figure 3).

An earlier version cut a tube around each track out of the image and shared charge at crossings with an EM step of its own; its tubes were twice as wide as the tracks (its width estimate collected neighbours and halos), it took X-rays lying on the muon's path (5–8 % of the Fe-55 events on L2) and left crumbs. The hint version has none of these.

2.4 Muon lines (default since 29 October)

With the hints, a muon's cluster is the set of clouds on its axis; clouds there also carry charge of crossing tracks, neighbours and X-rays, so the muon took 25–40 % foreign charge. With `crowded_shapes='on'` (`clustering_tests/crowded_shapes.py`) each hinted track is instead one component of the mixture with a physical shape,

$$L(p) = a_j(t)\; e(t)\; P\big(d;\, \sigma(t)\big),$$

t, d the pixel's position along / across the axis (the axis is refined: a lateral offset at each end), a_j the charge per px of length in 8-px segments (free: Landau fluctuations, knots), $\sigma(t)$ the width, linear along the track (diffusion grows with depth) and capped at 1.2× the tracker's rms within 2–3 px, P the pixel-integrated Gaussian and $e(t)$ soft ends. The clouds A_GMX placed on the axis are handed to the line; clouds and lines are refitted together (EM of the clouds with the lines as a fixed part of the model, each line refitted from its responsibility share), compact charge they do not explain is re-seeded, clouds centred on a line with the line's own width are its knots (a clearly sharper or wider cloud on the path, e.g. a front-side X-ray, stays separate), and A_GMX's final stage (`gmx.general_cluster(mix0=, fixed=, post_lab=)`) assigns everything else. Every pixel's charge is split between lines and clouds by model responsibility, so a touching track keeps its clouds and its charge, and the clusters do not grow into a line's footprint.

2.5 Topology split (default since 29 October)

One physical event has a simple skeleton: an X-ray is round, a muon or electron track a path. On the minimum spanning tree of each cluster's cloud centres: where ≥ 3 branches of ≥ 10 px meet, the two that continue each other most straightly stay together and the others become separate clusters (short spurs stay); two compact parts (≤ 15 px, each ≥ 20 % of the charge) whose model density (smoothed 0.8 px) dips below 0.35 between their centres are cut apart; a compact round X-ray-like cloud (1–22 keV, $\sigma \leq 1$ px, axis ratio < 1.5) attached to a larger group through a dip is split off.

Tried and dropped. Curly electron tracks as *path* components (a spine through the cluster's clouds, charge per length and width fitted, fragments joined end to end) lowered every score (electron tracks whole 71 → 70 %, muons "correct" 43.5 → 40 %, Fe-55 events −3 %); a one-or-two round Gaussian refit of X-ray clusters changed nothing on the injected pairs. Neither is in the code.

2.6 Crossing split

After the merge graph and after the island arbitration, a cluster whose clouds hold two straight lines (RANSAC on the cloud centres, each ≥ 20 px and ≥ 3 clouds) that cross as an X ($\geq 20°$, both continuing ≥ 8 px past the crossing) is split into the two tracks; clouds go to the nearer line.

2.7 Track linking

Collinear straight fragments of one track are joined (both pieces $\geq$ 12 px, endpoints within $\max(2.5,\, 1.5 + 0.01L)$ px of the longer piece's axis, parallel within 4°, gap < 15 px, or < 25 px for pieces ≥ 30 px). Shorter pieces — photo-electron tracks of 59.5-keV gammas, X-rays — are never joined (with 5 px, Am-241 EXT3 lost 4 % of its 59.5-keV events). Streaks are not linked.

3 Results

3.1 L2 surface run: maps

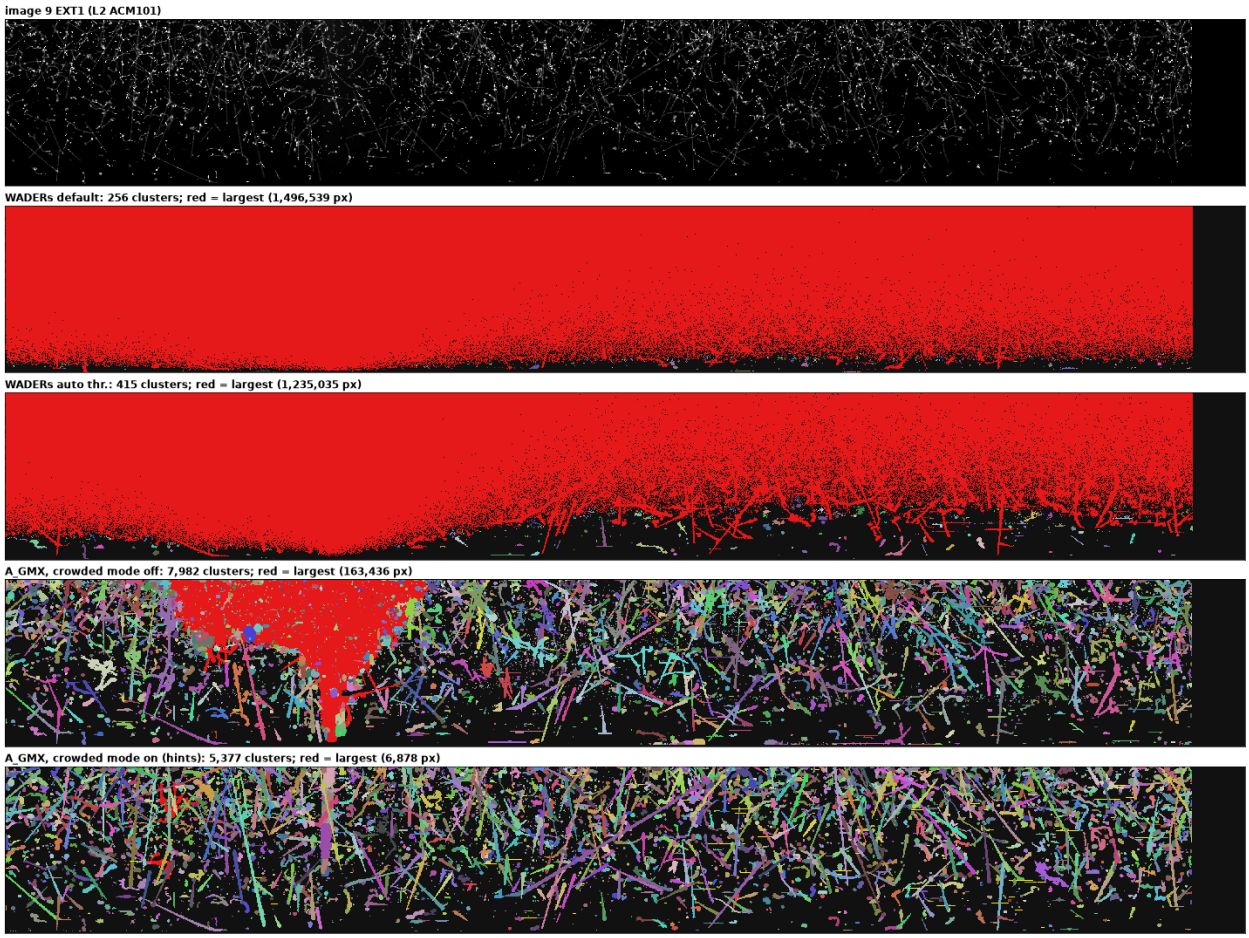


Figure 1: L2 image 9 EXT1. From the top: image; WADERs (default and electron thresholds: one cluster covers most of the CCD); A_GMX without and with crowded mode. Red: the largest cluster; yellow: serial-register streaks.

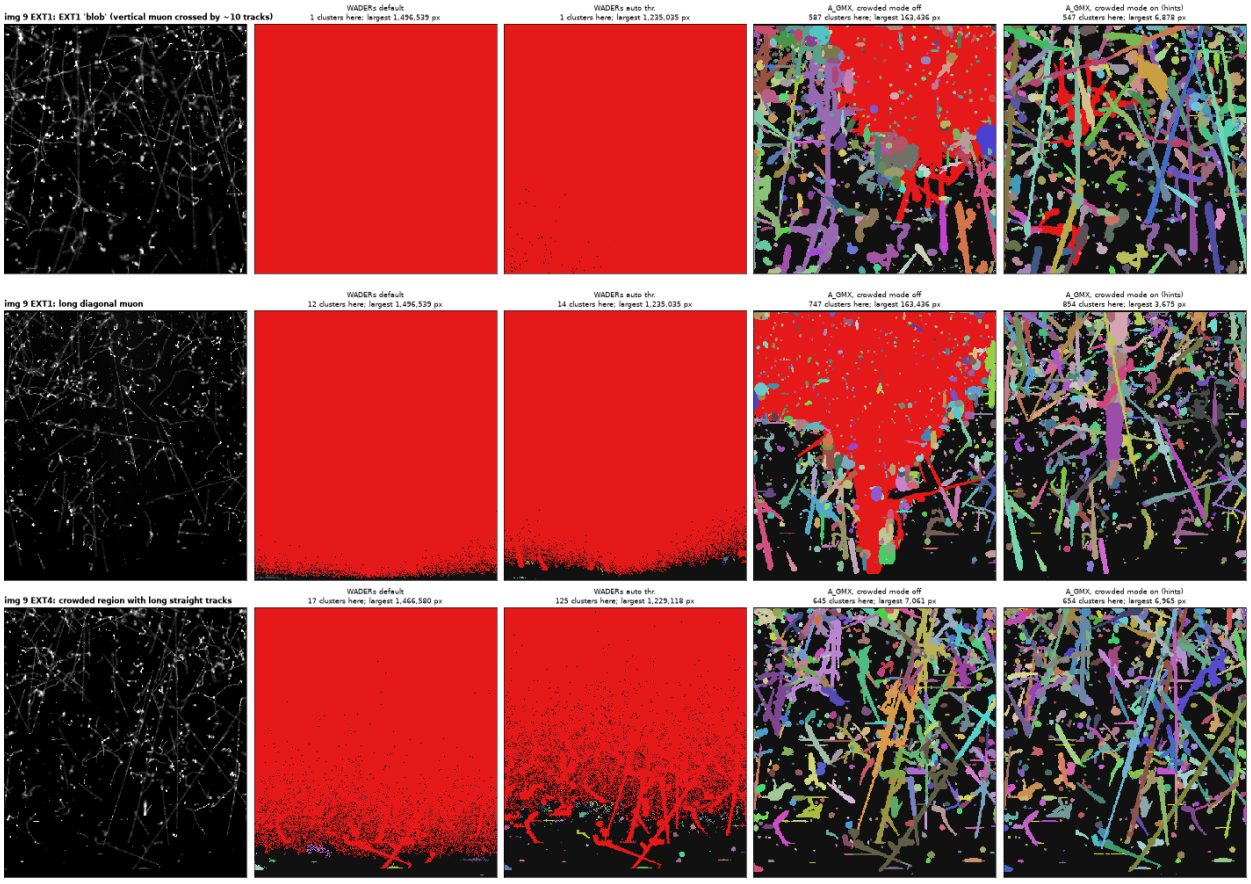


Figure 2: Close-ups: the EXT1 "blob" (a vertical muon crossed by about ten tracks), a long diagonal muon, a crowded region of EXT4. Columns as above; titles give the number of clusters in the cut-out and the largest.

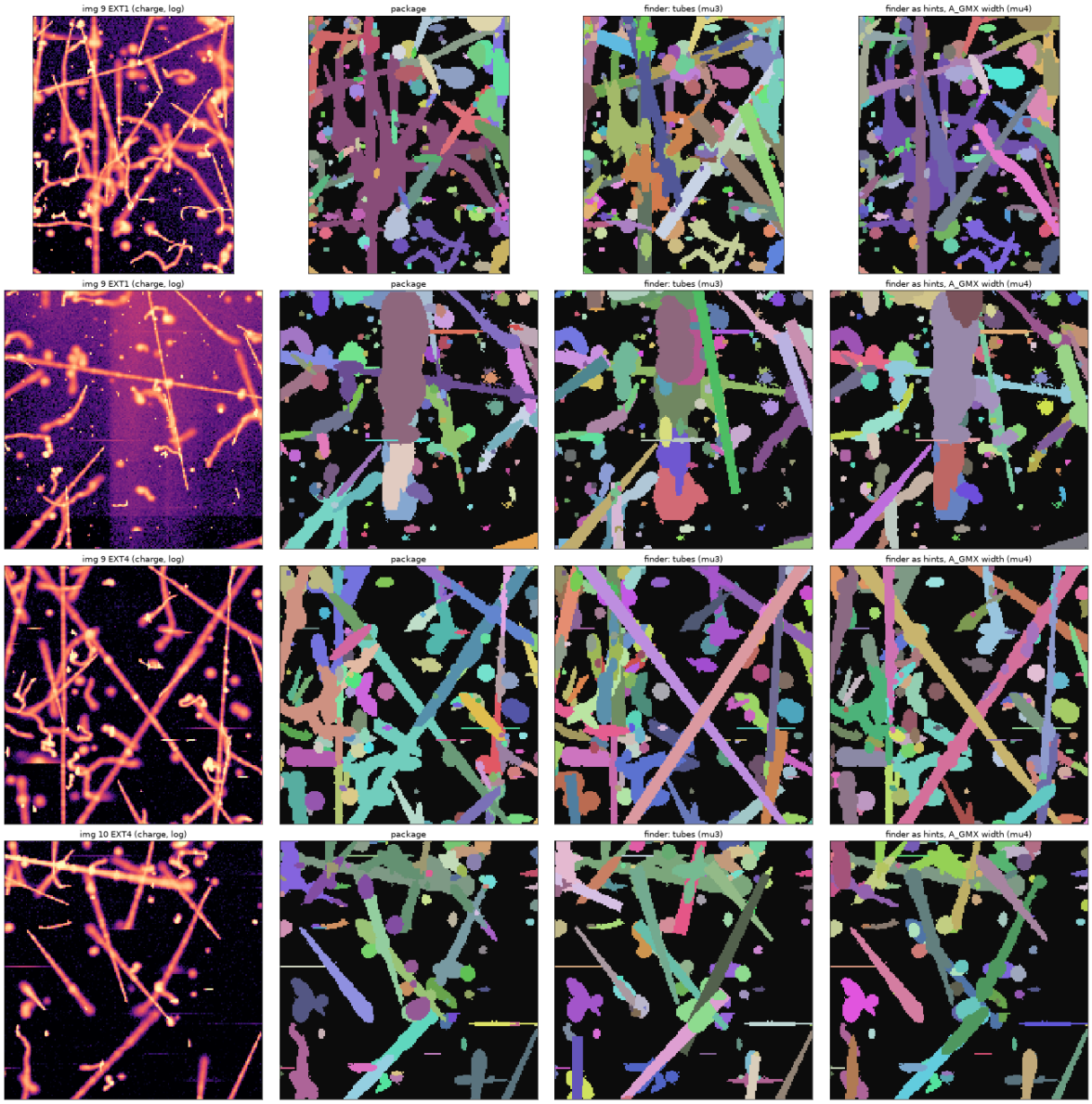


Figure 3: Track width. Columns: charge; the package without straight tracks; the earlier tube version (too wide); the hint version, where the width is A_GMX's own.

3.2 L2: injected muons

25 muons per image (projected length 40–400 px, Landau-fluctuating charge, diffusion growing from 0.2 to 0.8–1.6 px with depth) were added to the real L2 images 9 and 10 (EXT1, EXT4; 2 seeds: 200 muons) and the images clustered.

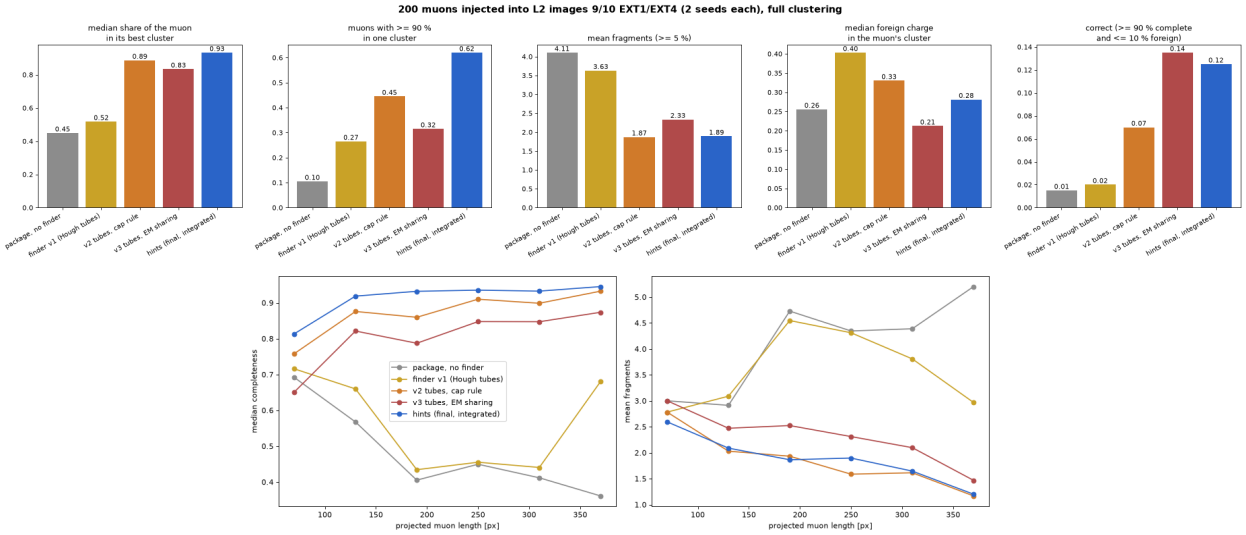


Figure 4: Per muon: share of its charge in its best cluster, fraction with $\geq 90\,\%$ in one cluster, number of fragments ($\geq 5\,\%$), foreign charge in its cluster, and "correct" ($\geq 90\,\%$ in one cluster and $\leq 10\,\%$ foreign). The last bar is the final, integrated code. In this image a 300-px line crosses about 15 other tracks, so few muons can be "correct".

	no straight tracks	tubes (earlier)	hints (final)
median share in the best cluster	0.45	0.83	0.93
$\geq 90\,\%$ in one cluster	10 %	32 %	62 %
fragments per muon	4.1	2.3	1.9

3.3 L2: Fe-55

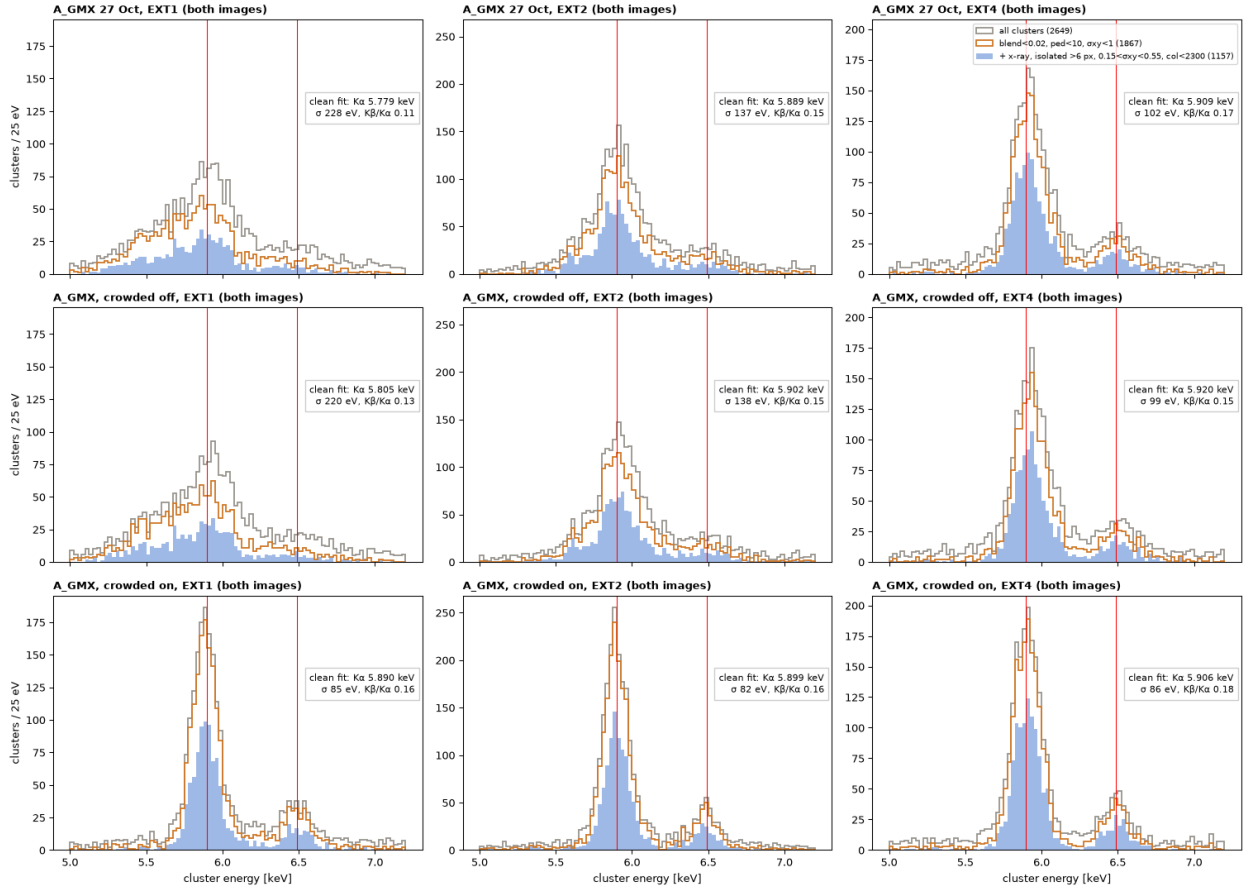


Figure 5: Fe-55 on L2 (images 9 and 10), front side. Rows: A_GMX of 27 October; today's A_GMX with crowded mode off; with crowded mode on. Blue: clean selection (blend < 0.02, ped < 10, $\sigma_{xy} < 1$, x-ray class, isolated > 6 px, $0.15 < \sigma_{xy} < 0.55$, column < 2300). Energies from 29 October use $\varepsilon(130\,\mathrm{K}) = 3.7486\,\mathrm{eV}$.

clean Kα events / width	EXT1	EXT2	EXT4
crowded off	476 / 220 eV	797 / 138 eV	896 / 99 eV
crowded on	785 / 85 eV	1016 / 82 eV	1021 / 86 eV

3.4 Am-241, EXT4 (the crowded CCD)

Ten of the 29 AMTest images (every third) were reclustered with crowded mode on and compared with the A_GMX clusters of 27 October on the same images (energies scaled to the same ε).

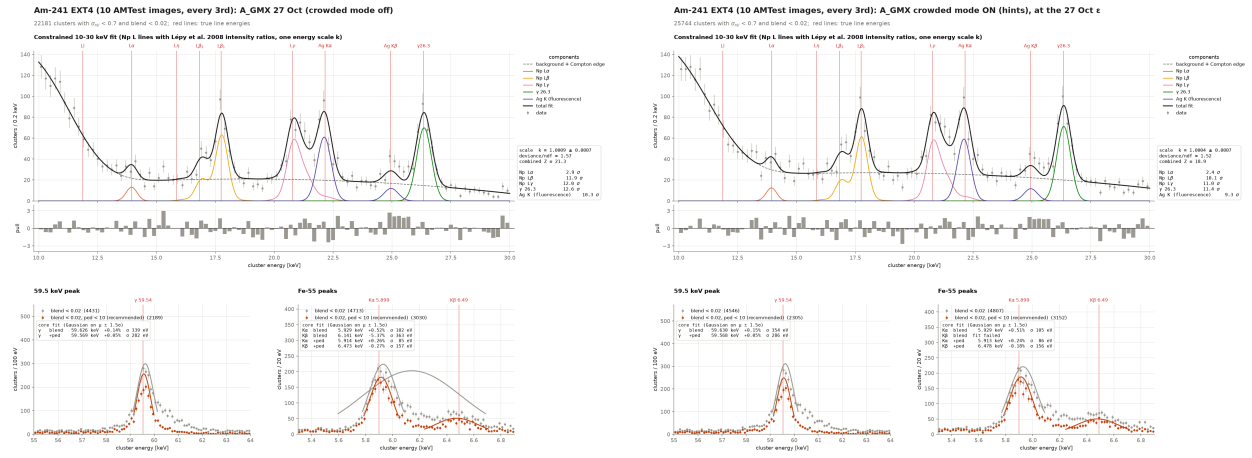


Figure 6: Constrained 10–30 keV fit (Np L lines with Lépy et al. intensity ratios, one energy scale k), 59.5-keV and Fe-55 peaks. Left: without crowded mode ($k = 1.0009 \pm 0.0007$, $Z = 21.2$, 22 181 clusters); right: with crowded mode ($k = 1.0004 \pm 0.0007$, $Z = 19.3$, 25 744 clusters).

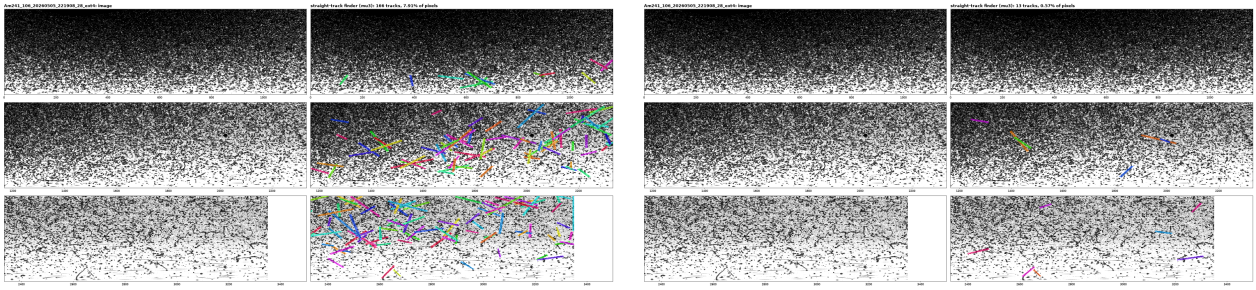


Figure 7: Straight tracks on an Am-241 EXT4 image without (left, 166) and with (right, 13) the continuity test.

3.5 Am-241, EXT3

With crowded mode the fit routine (`amfit.Model.fit`, 9 starts) first returned broad lines at the upper bound of the width ($s_0 = 0.8$ keV, $k = 0.973$, $Z = 4.7$). That was a local minimum: starting from narrow widths held fixed and then freed gives a likelihood higher by $\Delta \ln L = 42$ with $s_0 = 0.05$ keV, the same as without crowded mode. The figures use this extended start. Crowded mode adds 7–12 % events to the Np L lines and 26.3 keV and removes 2.8 % at 59.5 keV (with the 59.5-keV width 21 eV narrower).

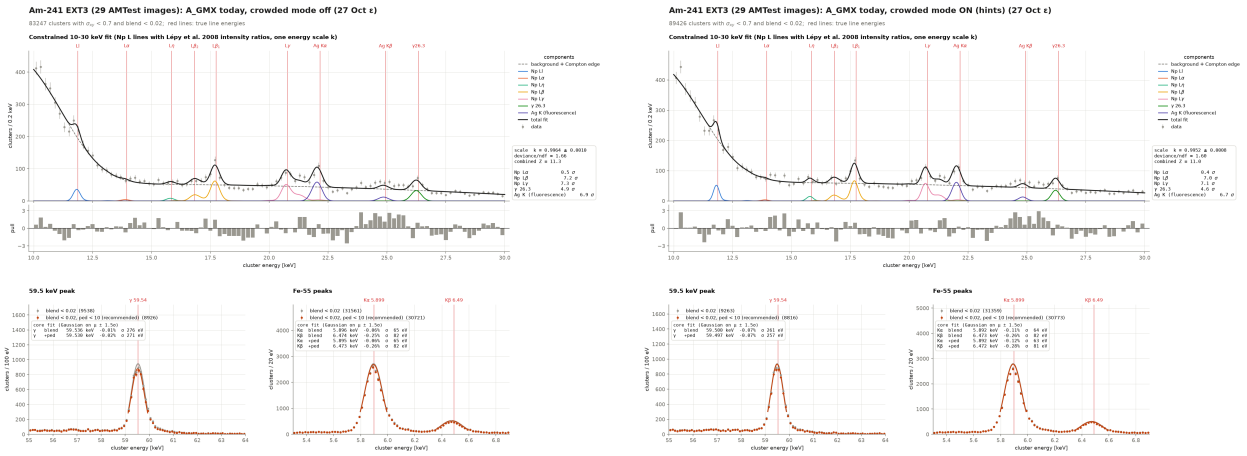


Figure 8: EXT3, 29 images, today's code with crowded mode off (left: $k = 0.9964 \pm 0.0010$, deviance/ndf 1.66, $Z = 11.3$) and on (right: $k = 0.9952 \pm 0.0008$, 1.60, $Z = 11.0$). The 59.5-keV and Fe-55 peaks move by 3–33 eV.

3.6 Muon lines and topology: injections and the muon energy

L2 injections (study version)	hints	lines + topology
muons: median foreign charge in the muon's cluster	0.28	**0.085**
muons: $> 30\,\%$ foreign	47 %	**10 %**
muons: $\geq 90\,\%$ in one cluster	62 %	56 %
muons: "correct" ($\geq 90\,\%$ in one, $\leq 10\,\%$ foreign)	12.5 %	**43.5 %**
electron tracks (50–500 keV): $\geq 90\,\%$ in one cluster	66 %	**71 %**
electron tracks: median foreign charge	0.20	**0.056**
touching Fe-55 X-ray pairs (2.5–7 px) resolved	72 %	**76 %**
Fe-55 clean Kα events (L2, 3 CCDs)	2849	2758 (−3 %)

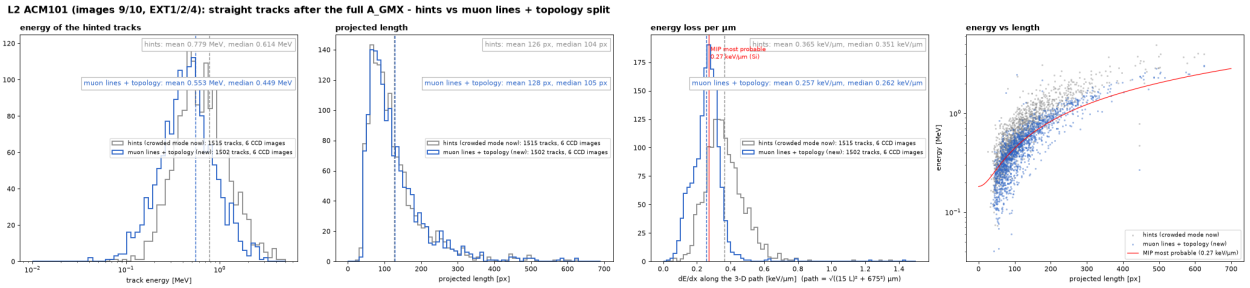


Figure 9: Straight tracks on L2 (6 CCDs, ∼1500 tracks) after the full A_GMX: hints (grey) and muon lines with the topology split (blue). With the hints the energy loss along the 3-D path peaks at 0.34 keV/µm, 25 % above the most probable value of a minimum-ionising particle in 675 µm of silicon (0.27–0.29 keV/µm), because the clusters carry charge of the tracks they cross; with the lines it peaks at 0.27 keV/µm and the mean energy drops from 0.78 to 0.55 MeV.

3.7 Energy of the straight tracks

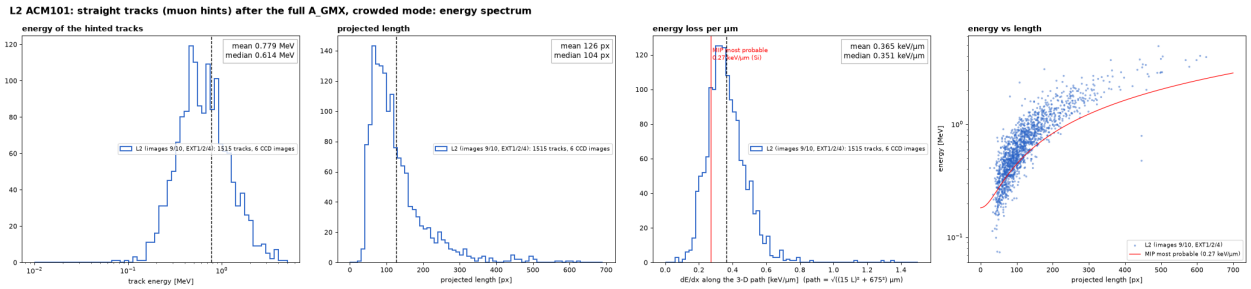


Figure 10: Clusters defined by the hints: energy, projected length, energy loss along the 3-D path $\sqrt{(15L)^2 + 675^2}$ µm and energy vs length, L2 ACM101. Over 1515 tracks the mean energy is 0.78 MeV (median 0.61 MeV) and the most probable energy loss 0.34 keV/µm, ∼25 % above the 0.27 keV/µm of a minimum-ionising particle: the clusters also carry their share of the tracks they cross, and some are straight stretches of electron tracks.

4 Open items

- **Am-241 EXT3:** 59.5 keV loses 2.8 % of its events with crowded mode (not traced to one piece).
- **Fit routine:** `amfit.Model.fit` can end in a wide-line minimum; it should also start from narrow widths.
- **Electron tracks:** straight 50–90 px stretches of curly electron tracks are taken as straight tracks.
- **Muon lines:** slightly fewer muons reach 90 % completeness (62 → 56 %) and the median energy loss (0.262) sits just below the most probable value: a few percent of a muon's charge goes to neighbours at crossings and knots.
- **Topology split:** long curly electron tracks (> 300 keV) are split a little more often; Fe-55 clean events −3 %.
- **Validation:** the injected muons are simulated (Moyal fluctuations, linear diffusion); EXT4 was checked on 10 of 29 images.

5 How to run

```
run_WADERs images.txt --gain-correction --clusterer agmx --agmx-crowded auto # crowded
    mode where the core fill >= 0.10
run_AGMX WADERs_analysis_<run> --agmx-crowded on # force it on a finished run
run_AGMX WADERs_analysis_<run> --agmx-crowded-shapes off # crowded mode with hints, no
    lines / topology
AGMX_CROWDED=auto python my_script.py # in python: environment default
```

```
import adaptive as AD
clusters, info = AD.adaptive_cluster(Q, valid, engine="gmx", crowded="auto", return_info=
    True)
info["core_fill"], info["crowded_mode"], info["crowded_shapes"], info["n_lines"], info["
    n_topology_splits"], info["n_streaks"]
```

Each output file records the setting and what ran in `agmx_config` (key `crowded_mode`). Code: `clustering_tests/crowded.py` (trigger, background, streaks, hints, crossing split, linking), `crowded_shapes.py` (muon lines, topology), `adaptive.py` (the switches), `gmx.py` (`general_cluster(split_crossings=, hints=, mix0=, fixed=, post_lab=)`); tests: `tests/test_agmx_cro` the study that led here: `wip/agmx_crowded/`.